

SVKM'S NMIMS
MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING
SCHOOL OF TECHNOLOGY MANAGEMENT

Academic Year: 2023-2024



Program/s: B. Tech

Year: II Semester: IV

Stream/s : CSE (Cybersecurity)

Subject: Introduction to Cryptography

Time: 3 hrs (10:00 am. To 01:00 pm.)

Date: 22 / 04 / 2024

No. of Pages: 03

Marks: 100

Final Examination/ Re-Examination (2021-22,2022-23)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Question No. 1 is compulsory.
- 2) Out of remaining questions, attempt any 4 questions.
- 3) In all 5 questions to be attempted.
- 4) All questions carry equal marks.
- 5) Answer to each new question to be started on a fresh page.
- 6) Figures in brackets on the right hand side indicate full marks.
- 7) Assume Suitable data if necessary.

Q 1	Solve:	
CO - 1; BL- 2	a. Ciphertext given below is obtained by first encrypting the plaintext by using Auto key cipher with the key 9 and then encrypting the result again by the permutation cipher with the key: $\pi = \begin{array}{ c c c c c } \hline 1 & 2 & 3 & 4 & 5 \\ \hline 5 & 4 & 1 & 3 & 2 \\ \hline \end{array}$ Decrypt the text: VZBGHEVTSHVZKJY	[05]
CO - 1; BL- 2	b. Suppose that the message "brick" is encrypted as CBKRI by using the permutation cipher with a key π. Write down the key by filling out the missing entries in the permutation π: $\begin{array}{ c c c c c } \hline 1 & 2 & 3 & 4 & 5 \\ \hline & & & & \\ \hline \end{array}$	[05]
CO - 1; BL- 2	c. You found one paragraph ciphertext that was encrypted by using Vigenere cipher. You found that the string: BQJTWDAQZALB occurs many times. Suppose you make a guess correctly that the repeating above ciphertext is the encryption of the plaintext "cryptography". If you know that the keyword is a 7 letters word, find the key word. <i>Your answer should be a meaningful word with 7 letters.</i>	[05]
CO - 1; BL- 2	d. Following cipher text was encrypted by a Shift Cipher mod 26: WTQP TD L RTQE Decrypt the message by finding the key. <i>Hint: There is a one letter word.</i>	[05]

Q 2 CO - 2; BL- 2	a. Compute $\lambda(55)$ where $\lambda(n) = \frac{\phi(n)}{\gcd(p-1, q-1)}$ Recall that a modified RSA cryptosystem is defined similarly as regular RSA system except that a is calculated by requiring $ab \equiv 1 \pmod{\lambda(n)}$ instead of $ab \equiv 1 \pmod{\phi(n)}$. Compute a for the modified RSA cryptosystem with $(55, 3)$ as the public key.	[10]																																																																								
Q 2 CO - 2; BL- 2	b. Bob accidentally selected n such that $n + 9^2$ is a perfect square, that is, $n + 9^2 = s^2$ for some positive integer s. Find the factors of n. <i>Hint:</i> $a^2 - b^2 = (a + b)(a - b)$. Also find factors of $n = 2189284635403183$.	[10]																																																																								
Q 3 CO - 2; BL- 4	a. Let M be the Multiplicative Cipher where the encryption rule is given by $e_k(x) = kx \pmod{26}$ and S be the Shift Cipher. Consider the key $(5, 12)$ in $M \times S$. Find the corresponding key in $S \times M$ that gives the same encryption rule.	[10]																																																																								
Q 3 CO - 1; BL- 2	b. Notice that the key stream 1101010110101011 of a LFSR has period seven. Find the degree of this key stream. Suppose that the first eight bits of the key stream of a LFRS of degree 3 is 10100111. Find next four bits of this key stream.	[10]																																																																								
Q 4 CO - 1; BL- 6	a. Compute $N_L(6, 4)$ by filling out the following table: <table><tr><td>X_1</td><td>X_2</td><td>X_3</td><td>Y_1</td><td>Y_2</td><td>Y_3</td><td>6</td><td>4</td></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td></td><td></td><td></td><td></td></tr></table> Compute the bias $\epsilon(6, 4)$. The pair $(6, 4)$ can be used to construct linear approximation for the given SPN. Is it True or False?	X_1	X_2	X_3	Y_1	Y_2	Y_3	6	4	0	0	0						0	0	1						0	1	0						0	1	1						1	0	0						1	0	1						1	1	0						1	1	1						[10]
X_1	X_2	X_3	Y_1	Y_2	Y_3	6	4																																																																			
0	0	0																																																																								
0	0	1																																																																								
0	1	0																																																																								
0	1	1																																																																								
1	0	0																																																																								
1	0	1																																																																								
1	1	0																																																																								
1	1	1																																																																								
Q 4 CO - 1; BL- 2	b. Given the input difference $x' = 101$, fill out the following table to find the corresponding output difference distribution by using $x^* = x \oplus x'$, $y^* = \pi_S(x^*)$ and $y \oplus y^* = y'$: <table><tr><td>x</td><td>x^*</td><td>y</td><td>y^*</td><td>y'</td></tr><tr><td>000</td><td></td><td>110</td><td></td><td></td></tr><tr><td>001</td><td></td><td>101</td><td></td><td></td></tr><tr><td>010</td><td></td><td>001</td><td></td><td></td></tr><tr><td>011</td><td></td><td>000</td><td></td><td></td></tr><tr><td>100</td><td></td><td>011</td><td></td><td></td></tr><tr><td>101</td><td></td><td>010</td><td></td><td></td></tr><tr><td>110</td><td></td><td>111</td><td></td><td></td></tr><tr><td>111</td><td></td><td>100</td><td></td><td></td></tr></table>	x	x^*	y	y^*	y'	000		110			001		101			010		001			011		000			100		011			101		010			110		111			111		100			[10]																											
x	x^*	y	y^*	y'																																																																						
000		110																																																																								
001		101																																																																								
010		001																																																																								
011		000																																																																								
100		011																																																																								
101		010																																																																								
110		111																																																																								
111		100																																																																								
Q 5 CO - 2; BL- 6	a. By looking at the last column of the above table <i>Q.4b</i>, find the distribution for the corresponding output difference: <table><tr><td>y'</td><td>000</td><td>001</td><td>010</td><td>011</td><td>100</td><td>101</td><td>110</td><td>111</td></tr><tr><td>freq.</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>	y'	000	001	010	011	100	101	110	111	freq.									[10]																																																						
y'	000	001	010	011	100	101	110	111																																																																		
freq.																																																																										

		Find all b such that $N_D(9, b)$ is the maximum value. Compute the corresponding $R_p(9, b)$																	
Q 5 CO - 1; BL- 2	b.	Solve $12x + 4 = 14 \pmod{26}$ and find both the solutions by using the fact that $6(11) = 66 = 5(13) + 1$	[10]																
Q 6 CO - 1; BL- 3	a.	Consider the second S-box for SDES which is given below: <table border="1" style="margin: 10px auto;"><tr><td>5</td><td>2</td><td>1</td><td>6</td><td>3</td><td>4</td><td>7</td><td>0</td></tr><tr><td>1</td><td>4</td><td>6</td><td>2</td><td>0</td><td>7</td><td>5</td><td>3</td></tr></table> Recall that the first bit of the input decides the row and the last three bits decide the column of the S-box. If the input is 1011, find the corresponding output given by the above S-box.	5	2	1	6	3	4	7	0	1	4	6	2	0	7	5	3	[10]
5	2	1	6	3	4	7	0												
1	4	6	2	0	7	5	3												
Q 6 CO - 1; BL- 3	b.	Find all the inputs for the above S-box for SDES cipher which yield the output 101	[10]																
Q 7 CO - 1; BL- 3	a.	$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}$ Find the inverse of the following matrix in Z_{26} :	[10]																
Q 7 CO - 1; BL- 3	b.	Complete the division algorithm table to find the gcd (17, 5). Find the value of x and y such that $17x + 5y = \text{gcd}(17, 5)$. Find the inverse of 5 in mod 17.	[10]																

Alphabet Codings

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

Inverses in Z_{26}

a	1	3	5	7	11	17	25
a^{-1}	1	9	21	15	19	23	25

S-box π_S for a simple SPN

input	000	001	010	011	100	101	110	111
output	110	101	001	000	011	010	111	100